Summary Report (Literature review)

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Topic- ***From Self-excitation to Saturation: The Riga Dynamo Story***

The *Riga Dynamo Experiment* provides the first laboratory verification of homogeneous *hydromagnetic self-excitation* in liquid sodium. The model of the dynamo was inspired from *Ponomarenko*'s theoretical helical screw flow model with a few modifications. [1][4]

It combines two mathematical frameworks-

1. *The Magnetohydrodynamic Induction Equation*

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) + \lambda \nabla^2 \mathbf{B}$$

2. *Navier-Stokes Momentum Equation*

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} = -\frac{\nabla p}{\rho} + \frac{1}{\mu_0 \rho}(\nabla \times \mathbf{B}) \times \mathbf{B} + \nu \nabla^2 \mathbf{v} + \mathbf{f}_{\text{drive}}$$

where rotating magnetic field eigenmodes grow as $\mathbf{B}(t) \propto e^{(p+i2\pi f)t}$ (calculated from 1), until saturation kicks in. The experiment can be divided into multiple regimes – each being a range of rotation per minute (rpm) values with their own growth rate (p) of magnetic field. [1]

The *first phase* (November 1999) of the experiment examines the *kinematic regime*, where the resultant Lorentz force due to the magnetic field was not strong enough to trigger saturation. In the linear kinematic regime, *subcritical* fluid rotation (1980 rpm) yields *exponential field decay* (p = -0.3 s^{-1} and f = 1.1 Hz), whereas crossing the critical magnetic Reynolds number (R_{mc}) triggers *exponential growth of a 3D non-axisymmetric rotating* (m=1) eigenmode (p = 0.0315 s^{-1} and f = 1.326Hz) above a certain *supercritical* rotation (2150 rpm). [1][2]

In the *second phase* (July 2000), in *non-linear saturation regime*, net growth ceases (p=0) as the *azimuthal components of Lorentz forces* become large enough, exerting a dynamic back-reaction on fluid velocity and inducing *localized flow deformation*, braking rotation (-1.2 m/s) near the cylinder bottom while accelerating return flow near the top (0.6 m/s). This velocity redistribution shifts the *magnetic field envelope upward* toward the propeller while keeping oscillation frequencies, following the same trend as that of kinematic regime. [1][3]

References

[1] Gailitis, A., Lielausis, O., Platacis, E. *et al.* The Riga Dynamo Experiment. *Surveys in Geophysics* 24, 247–267 (2003).

[2] Gailitis, A., Lielausis, O., Dement'ev, S., Platacis, E., Cifersons, A., Gerbeth, G., Gundrum, Th., Stefani, F., Christen, M., Hänel, H., and Will, G.: 2000, 'Detection of a Flow Induced Magnetic Field Eigenmode in the Riga Dynamo Facility', Phys.Rev.Lett.84, 4365–4368.

[3] Gailitis, A., Lielausis, O., Platacis, E., Dement'ev, S., Cifersons, A., Gerbeth, G., Gundrum, Th., Stefani, F., Christen, M., and Will, G.: 2001a, 'Magnetic Field Saturation in the Riga Dynamo Experiment', Phys. Rev. Lett. 86, 3024–3027.

[4] Ponomarenko, Yu. B.: 1973, 'On the Theory of Hydromagnetic Dynamo', J. Appl. Mech. Tech. Phys. 14, 775–779.